

# The rules of differentiation

Lessons 7–9 · Algebra 11, §1.4 · P1 · 7.2–7.4 · 28 problems

Differentiate with respect to  $x$ . No chain rule is needed anywhere on this sheet.

## Easy 7

One rule applied once. Problems 6 and 7 are the test: the expression has to be rewritten as an index before the power rule is usable.

1  $(x^9)'$   $9x^8$

Power rule: the 9 comes down, the index drops by one.

2  $(7x^3)'$   $21x^2$

Constant multiple:  $7 \times 3x^2 = 21x^2$ .

3  $(x^5 - x^2)'$   $5x^4 - 2x$

Differentiate term by term.

4  $(-12)'$   $0$

A constant does not change, so its rate of change is 0.

5  $(6x + 1)'$   $6$

A straight line of gradient 6; the +1 contributes nothing.

6  $(\frac{1}{x^3})'$   $-\frac{3}{x^4}$

Rewrite as  $x^{-3}$ . The power rule gives  $-3x^{-4} = -\frac{3}{x^4}$ .

7  $(2\sqrt{x})'$   $\frac{1}{\sqrt{x}}$

Rewrite as  $2x^{1/2}$ . Then  $2 \times \frac{1}{2}x^{-1/2} = x^{-1/2} = \frac{1}{\sqrt{x}}$ .

## Medium 7

Choosing the shorter route. In 11, 12 and 14 expanding or splitting beats the product and quotient rules; 13 genuinely needs the quotient rule.

8  $(5x^3 + 2x^2 - 8x + 1)'$   $15x^2 + 4x - 8$

Term by term; the +1 vanishes.

9  $(\frac{4}{x^5})'$   $-\frac{20}{x^6}$

$\frac{4}{x^5} = 4x^{-5}$ , so the derivative is  $-20x^{-6}$ .

10  $(x^2\sqrt{x})' \quad \left| \quad \frac{5}{2}x^{3/2} \right|$

$x^2\sqrt{x} = x^2 \cdot x^{1/2} = x^{5/2}$ , so the derivative is  $\frac{5}{2}x^{3/2}$ .

11  $((x-2)(x+6))' \quad \left| \quad 2x+4 \right|$

Expanding is quicker:  $(x-2)(x+6) = x^2 + 4x - 12$ .

12  $(3x(x^2-4))' \quad \left| \quad 9x^2-12 \right|$

Expand first:  $3x^3 - 12x$ .

13  $\left(\frac{x+3}{x-1}\right)' \quad \left| \quad -\frac{4}{(x-1)^2} \right|$

Quotient rule with  $u = x+3$ ,  $v = x-1$ . The numerator is  $1 \cdot (x-1) - (x+3) \cdot 1 = -4$ .

14  $\left(\frac{x^3-5x}{x}\right)' \quad \left| \quad 2x \right|$

Divide first:  $\frac{x^3-5x}{x} = x^2 - 5$ , whose derivative is  $2x$ .

## Hard 7

*Method and condition. In 16 the numerator must collapse to a constant; 17 and 18 want index form; 21 gives the gradient and asks for the point, which reverses the usual direction.*

15  $((3x+2)(x^2-x))' \quad \left| \quad 9x^2-2x-2 \right|$

Product rule with  $u = 3x+2$ ,  $v = x^2-x$ :  $3(x^2-x) + (3x+2)(2x-1) = 3x^2 - 3x + 6x^2 + x - 2 = 9x^2 - 2x - 2$ . Expanding to  $3x^3 - x^2 - 2x$  first gives the same answer.

16  $\left(\frac{x^2+1}{x-3}\right)' \quad \left| \quad \frac{x^2-6x-1}{(x-3)^2} \right|$

Quotient rule with  $u = x^2+1$ ,  $v = x-3$ . The numerator is  $2x(x-3) - (x^2+1) = 2x^2 - 6x - x^2 - 1 = x^2 - 6x - 1$ .

17  $(x^4\sqrt{x})' \quad \left| \quad \frac{9}{2}x^{7/2} \right|$

$x^4\sqrt{x} = x^{9/2}$ , so the derivative is  $\frac{9}{2}x^{7/2}$ .

18  $\left(\frac{x+4}{\sqrt{x}}\right)' \quad \left| \quad \frac{1}{2\sqrt{x}} - \frac{2}{x\sqrt{x}} \right|$

Split the fraction rather than using the quotient rule:  $\frac{x+4}{\sqrt{x}} = x^{1/2} + 4x^{-1/2}$ . Differentiating gives  $\frac{1}{2}x^{-1/2} - 2x^{-3/2}$ .

19  $(x^3(x-2)(x+2))' \quad \left| \quad 5x^4-12x^2 \right|$

Recognise the difference of two squares first:  $x^3(x^2-4) = x^5 - 4x^3$ , so the derivative is  $5x^4 - 12x^2$ .

20 Find  $y''$  for  $y = 2x^5 - x^3 + 4x \quad \left| \quad 40x^3 - 6x \right|$

$y' = 10x^4 - 3x^2 + 4$ , and differentiating again gives  $y'' = 40x^3 - 6x$ .

- 21 Find the values of  $x$  at which  $y = x^3 - 12x$  has gradient 15  $x = \pm 3$

Gradient means  $y'$ , so set  $3x^2 - 12 = 15$ . Then  $3x^2 = 27$ ,  $x^2 = 9$ , so  $x = 3$  or  $x = -3$ . A cubic reaches a given positive gradient twice.

### Very hard 7

Nothing here needs the chain rule. What is hard is choosing the form (22, 23, 26) or working backwards from a gradient to the function (27, 28).

- 22  $((2x - 1)(x + 1)(x + 3))' \quad 6x^2 + 14x + 2$

Expand rather than apply the product rule twice.  $(2x - 1)(x + 1) = 2x^2 + x - 1$ , and multiplying by  $(x + 3)$  gives  $2x^3 + 7x^2 + 2x - 3$ . The derivative is  $6x^2 + 14x + 2$ .

- 23  $(\frac{(x - 1)(x^2 + 4)}{x^2})' \quad 1 - \frac{4}{x^2} + \frac{8}{x^3}$

Expand the numerator to  $x^3 - x^2 + 4x - 4$ , then divide every term by  $x^2$ :  $x - 1 + 4x^{-1} - 4x^{-2}$ . Differentiating gives  $1 - 4x^{-2} + 8x^{-3}$ .

- 24 Find the gradient of  $y = \frac{2x + 1}{x^2 + 1}$  at  $x = 0$   $2$

Quotient rule. The numerator is  $2(x^2 + 1) - (2x + 1)(2x) = -2x^2 - 2x + 2$ , so  $y' = \frac{-2x^2 - 2x + 2}{(x^2 + 1)^2}$ . At  $x = 0$  this is  $\frac{2}{1} = 2$ .

- 25 Find the fourth derivative of  $y = \frac{1}{x^2}$   $\frac{120}{x^6}$

$y = x^{-2}$ . Then  $y' = -2x^{-3}$ ,  $y'' = 6x^{-4}$ ,  $y''' = -24x^{-5}$  and  $y^{(4)} = 120x^{-6}$ . The sign alternates and each coefficient multiplies by the next index.

- 26 Differentiate  $y = (x^2 - 3)^2$  **without** using the chain rule  $4x^3 - 12x$

Expand first:  $(x^2 - 3)^2 = x^4 - 6x^2 + 9$ . Differentiating term by term gives  $4x^3 - 12x$ .

- 27  $y = ax^2 + bx$  passes through (2, 2) and has gradient 5 at  $x = 2$ . Find  $a$  and  $b$ .  $a = 2, b = -3$

Two conditions, two unknowns. The point gives  $4a + 2b = 2$ , so  $2a + b = 1$ . The gradient  $y' = 2ax + b$  at  $x = 2$  gives  $4a + b = 5$ . Subtracting,  $2a = 4$ , so  $a = 2$  and  $b = -3$ . Check:  $y = 2x^2 - 3x$  gives  $y(2) = 2$  and  $y'(2) = 5$ .

- 28 The tangent to  $y = x^2 + \frac{k}{x}$  at  $x = 1$  is horizontal. Find  $k$ , and the value of  $y$  there.  $k = 2$ , and  $y = 3$

Horizontal means  $y' = 0$ . With  $y = x^2 + kx^{-1}$ ,  $y' = 2x - kx^{-2}$ , so at  $x = 1$ :  $2 - k = 0$ , giving  $k = 2$ . Then  $y(1) = 1 + 2 = 3$ .